

Simulated Fleet, Real Engine

The premise was that simulated robots need connecting to the Assignment Engine. Measured, the engine is not waiting on the simulator — it is waiting on **three** things, and the simulator is only one of them. Two of the other two are structural breaks no previous audit has recorded.

branch feature/dashboard HEAD 1223574 tree dirty suite 182 / 7 809 / 0 failures
migrations 29 this turn changed nothing

COORDINATOR INPUTS, TODAY 26 / 34 unresolved — measured live this turn	WITH EVERY CODE SEAM INJECTED 15 / 34 all REGISTER_UNRESOLVED — a new number	V1 STOP CONDITION 3 of 8 S-1, S-2, S-8 met — and must not move	BUILD GATES 7 / 1 PASS / FAIL — gate:composition
ROUTING READINESS BLOCKED D1, D3, D8 — and stays that way	SIMULATOR FIDELITY UNVALIDATABLE the correct, required result		

01 Executive summary

The milestone is achievable, and the existing Assignment Engine is the right and only path — but not for the reason the brief assumes.

Measured live this turn, read-only, via

```
coordinatorPipeline.requirements() against  
service.defaultSnapshot() :
```

```
== server.js as it stands today ==  
declared 34 · satisfied 8 · missing 26  
{ "EXTERNAL_ROUTING": 5, "REGISTER_UNRESOLVED": 15, "NO_PRODUCER": 6 }
```

```
== every code seam injected ==  
declared 34 · satisfied 19 · missing 15  
{ "REGISTER_UNRESOLVED": 15 }
```

That second line is the single most useful number in this report and it has not been recorded anywhere before. **Eleven of the twenty-six unresolved inputs are closed by code plus a declaration this milestone can legitimately produce.** The residual fifteen are all register calibration rows — a configuration publish act, two of them Safety-class that §22.3 forbids any automated process from choosing.

Three findings change the shape of the work

FINDING 1 · THE SECOND STRUCTURAL BREAK

`indexMaintainer.worker` is `DEFERRED` and **nothing starts it**. It is the only writer of `AgentCellPosition` and the Redis availability index, and *both* `candidates/expansion.candidatesInFineCell` and `coordinatorSolvePath.agentSnapshotLoaderFor` read them. With all 34 inputs supplied, candidate generation would return **zero agents** and every agent snapshot would carry a null position.

It is invisible to the requirements probe because the probe walks *constructor* seams and this is *a worker that is not scheduled*. It is repository-owned, and it is blocked by exactly two absent classifiers.

FINDING 2 · TWO MISSING PRODUCERS

No production code path creates `BatteryState` or `EnergyModelParams`. `batteryState.upsert` exists at one call site — `energyCalibration.worker.js:118`, a `DEFERRED` worker — plus `tools/verify/v1CorePath.js:158`. `energyModelParams.create` exists only in `tools/verify/phase9ProductionPath.js:155`. `robotSpecification.modelRowsFor` builds `EnergyModel` and **not** the §14.2 β coefficients.

So `agentSnapshotLoaderFor` yields `soc: null` — F7 and F34 refuse every candidate — and `consumption.legEnergyWh` returns `missing`:

["energyModelParams"] . Two more E-8b-family gaps: the column exists, the producer does not.

FINDING 3 · THE CHARGER IS THE MISSING CHARGING SCHEDULER

*The Scheduler is **outside this engine's boundary** (§1.6). It does not exist yet; blocking decision B2 records that a conformant publisher must be built or stubbed, and the engine must never compute a target SoC.*

src/engine/energy/chargingSchedulerClient.js, its own header

One charger, three plugs, FIFO, a target SoC, and a published availability projection is **that publisher**. Building it as a separate simulated service is not a demo shortcut — it is the architecturally correct home, and it simultaneously supplies F34/F35's inputs, the `chargingStatusFor` classifier that unblocks Finding 1, and `chargerReachability.worker`'s input. It is the highest-leverage single item in this plan.

One correction to the canonical boundary documents

V1_IMPLEMENTATION_CONTROL.md §9 row 8 and

coordinatorPipeline.js:410-412 both state "No mass column exists."

Robot.massKg exists — migration

20260906120000_robot_specification_and_chassis_class — documented in the schema as being there "because §14.2's consumption equation has a θ_{mass} term that needs one", with `robotSpecification.js` as its declared sole writer. The *value* is now operator-supplied at commissioning; only the `vehicleMassKgFor` **producer** is missing. That is a reclassification from

NO PRODUCER to **NEEDS INTEGRATION**.

WHAT MUST NOT BE CLAIMED — AND THIS PLAN DOES NOT CLAIM IT

Simulated evidence is not physical evidence. `sim:fidelity` stays UNVALIDATABLE. `routing:readiness` stays BLOCKED. `gate:composition` should stay RED. Phase 15's §24 table is untouched. **No stop condition should move, and S-3 cannot:** two of its fifteen register rows are Safety-class with no provisional route, and the region and charger declarations are owner acts.

02 Current architecture relevant to this milestone

Three layers. The milestone spans all three, and they are joined at exactly three verified points today.

LAYER	MODULES	STATE
Legacy operational plane	Robot / Task / Telemetry / Zone , dtaro.* , routing.service , mapbox.service , dashboard sockets	Live, w produc
Domain / engine plane (\$2 on)	Agent / AgentClass / Mission / Leg / Stop / Commitment / Outbox / WorkQueue , src/engine/** (290 modules)	Imple 10, 11 reache
Simulation plane	src/simulation/** (7 modules, 3 329 lines), simulator.* routes, SimulatedRobotPage	Live be ENABLE off by c

The three joins: the 1:1 Robot → Agent projection (Agent.robotDbId @unique); the \$10.3.1/\$11.2 wire protocol, which VirtualRobot already speaks in full; and — since Step 5 — positionObservation.service.js writing kind: "position" Observations from the telemetry path.

Stop-condition state, re-derived rather than copied: S-1 MET , S-2 MET , S-8 MET ; S-3 28 INPUTS , S-4 CONSEQUENCE , S-5 REFUSED — V9 + S2 , S-6 EXITS 1 , S-7 ONE HALF .

03 Simulation architecture audit

The protocol half is finished and excellent. The physics half diverges from the commissioned specification, silently.

COMPONENT	WHAT EXISTS	VERDICT
VirtualRobot.js 2 234 lines	Socket.IO client; AUTH; TELEMETRY with timestamp + sequence ; \$10.3.1 fencing via the shared engine/commitment/fencing ; \$11.5 durable dedup; \$11.2 OFFER_ACCEPT/REJECT/DEFER ; \$18.5	A · REUSE WHOLLY The spec's designated reference implementation. Do not fork

COMPONENT	WHAT EXISTS	VERDICT
	continuation limit; charging via the shared <code>chargeCurve</code>	
<code>SimulationEngine.js</code>	Lifecycle; re-reads the row and refuses a physical unit; per-robot seeded RNG; arbitrary robot count	A · REUSE
<code>simulationPolicy.js</code>	The <i>only</i> module evaluating the two conditions, 3 call sites	A · REUSE Do not add a fourth decider
<code>simulationConfig.js</code>	A closed nine -parameter tunable set; 13 <code>PROTECTED_FIELDS</code> refused by name	D · EXTEND THE CLOSED SET Not a config framework
<code>random.js</code>	mulberry32 + FNV-1a, two sub-streams per robot	A · REUSE
<code>constants.js</code>	<code>SPEED_BASE_MS = 5.56 ,</code> <code>PACK_NOMINAL_WH = 1000 , flat</code> <code>BATTERY_DRAIN_ACTIVE = 0.0444</code> <code>%/tick , CHARGE_PACK_TEMPERATURE_C</code> <code>= 20</code> — all fleet-wide	E · DEFECTIVE HERE
<code>POST</code> <code>/api/simulator/robot</code>	<code>requireElevatedRole()</code> ; server-generated id; requires a full specification — <code>chassisType</code> + 7 numbers; one robot per request, no limit	A · REUSE The path the presets plug into
§23.5 trust boundaries	<code>validatePosition</code> , <code>validateEnergy</code> , capability-claim rejection, persistent implausibility → quarantine + security event	A · REUSE Malformed-telemetry faults exercise a <i>real</i> control
<code>start / stop / reset</code> routes	Fleet-wide , <code>authUser</code> only — a documented limitation	SCOPE IS AN OWNER DECISION

3.1 The simulator's physics diverges from the commissioned specification

`robotSpecification.js` persists the operator's numbers onto `MobilityModel.speedModel.nominalSpeedMps` , `kinematicLimits.maxSpeedMps` , `EnergyModel.packNominalWh` , `ContainerModel.totalMassLimitKg` and `Robot.massKg` . **VirtualRobot** reads none of them.

- A robot commissioned at the owner's 1.2 m/s would be simulated at 5.56 m/s — 4.6× too fast, which is optimistic on travel time: the exact direction §24.4's one-sided fidelity gate exists to catch.
- The LIGHT / STANDARD / HEAVY distinction **would not exist in the simulation at all**. All three would move identically.
- A flat %/tick rate cannot meet the dynamic-energy requirement, and the module that *does* implement §14.2 — `consumption.legEnergyWh`, with `REQUIRED_PROFILE_FIELDS` — is already written and already refuses on absence.

A simulator that modelled the taper differently from the server would agree with it about every charging plan only by coincidence, and §24.4's fidelity gate would be measuring the difference between two models rather than between a model and reality.

`src/simulation/constants.js:39-54` — written about charging; now true of consumption

The same argument applies, and so does the same resolution: share the module.

3.2 One further defect, live the moment charging works

```
VirtualRobot._respondToOffer (:1181-1182)
  pathToPickup: stops[0]?.path || [],
  pathToDrop:   stops[1]?.path || [],
```

Exactly two stops. §13.4's `insertChargingStop` re-sequences and produces three or more. The third would be **silently dropped** and the agent would drive a truncated mission while reporting `TASK_COMPLETE`. Today unreachable; live the moment this milestone issues an offer against a low-battery agent. E · DEFECTIVE

04 Existing Mapbox / routing audit

Mapbox satisfies two of the six route fields, and cannot satisfy the production contract on grounds independent of field coverage.

CONTRACT FIELD	ENFORCED AT	MAPBOX TODAY	VERDICT
distanceM	<code>cellPairCache.buildEntry</code>	<code>route.distance</code>	REAL, AVAILABLE
travelSeconds	<code>cellPairCache.buildEntry</code>	<code>route.duration</code>	REAL, AVAILABLE
travelSdSeconds	<code>cellPairCache.buildEntry</code>	point estimate only	NO SOURCE N29 — no shortlisted engine returns a spread
climbM	<code>plan/timeline.project</code>	not requested	OBTAINABLE see \$05
descentM	<code>plan/timeline.project</code>	not requested	OBTAINABLE see \$05
stopStartCycles	<code>plan/timeline.project</code>	—	NO SOURCE from any Mapbox product
timeBucket	<code>cellPairCache.key</code>	nothing in <code>src/</code> produces one	NO PRODUCER
speedMetresPerSecond	<code>applyIntraCellOffset</code>	<i>not routing</i>	NOW AVAILABLE from <code>MobilityModel</code> per unit

It is also a hosted third-party API, and B1/R1 require a **declared, real, self-hosted** source. `routing:readiness` reports `BLOCKED` on D1/D3/D8 and **must** be left doing so.

The seam boundary is already correctly drawn

*It is **not** a routing source for the decision path... Presenting a Mapbox polyline as the missing traversal source would be exactly the fabrication the programme's standing rule forbids; drawing the line an already-chosen agent drives is not.*

`src/services/executionGeometry.service.js`, its own header

It also already solves a problem this milestone would otherwise hit: **no network call inside the commitment transaction** — geometry is resolved before `$transaction` opens, and absence means the stop carries *no path*,

which `assessExecutability` refuses with `NO_EXECUTABLE_PATH`. Both halves must be preserved. Reuse the precedent; do not cross it.

05 Terrain / elevation audit

The owner asked to inspect rather than assume. The answer has two halves and they disagree.

BACKEND — NO ELEVATION ANYWHERE

A search for `elevation|climbM|descentM|stopStartCycles|terrain` over `Backend/src` hits ten files, and **every one is a consumer or a declaration of absence**: the requirement, the seam, the cache that carries-when-present, and the two modules that refuse without it. **No producer. No DEM. No Tilequery. No elevation column in `schema.prisma`.**

FRONTEND — REAL ELEVATION, ALREADY LICENSED, ALREADY LOADED

```
export const TERRAIN = Object.freeze({
  sourceId: 'mapbox-dem',
  source: { type: 'raster-dem',
    url: 'mapbox://mapbox.mapbox-terrain-dem-v1',
    tileSize: 512, maxzoom: 14 },
  exaggeration: 1.0, // "the map visualizes truth (§24), and a s
                    // landscape would misrepresent gradients
                    // may be reading for route feasibility"
```

Real, vendor-measured elevation for RNSIT is **already in this product** — client-side, for rendering. So `climbM` / `descentM` for a development environment can come from a **measurement** — sampling the same DEM along the returned polyline, server-side — rather than from an invented number. That is a materially different act from fabrication, and the comment above is the repository's own statement that these gradients are treated as truth.

STILL HAS NO LEGITIMATE SOURCE — DO NOT INVENT

`stopStartCycles` — Mapbox `steps=true` returns maneuvers and intersections, which could *ground a derivation*, but a derivation is not a

measurement. `travelSdSeconds` — N29; no product returns it. Both are declared assumptions or absent.

Do not build a terrain engine. No `Elevation` table, no DEM cache service, no tile pipeline. A per-hop DEM sample behind the dev routing adapter, memoised by the cell pair `cellPairCache` already keys on, is the whole requirement.

06 **Battery / energy audit**

The schema is complete. It is also almost entirely unpopulated, and every engine module on this path refuses rather than defaults.

MODEL	FIELDS	PRODUCER IN <code>SRC</code> / ?
<code>EnergyModel</code>	<code>packNominalWh</code> , <code>chemistry</code> , <code>consumptionCoefficients</code> , <code>thermalDeratingCurve</code> , <code>chargePowerCurve</code>	<code>robotSpecification</code> writes the first two. The other three: none
<code>EnergyModelParams</code>	nine β coefficients, <code>betaThermal</code> , <code>betaPayloadThermal</code> , <code>stressCurves</code> , <code>residualCv</code> , <code>modelVersion</code>	NONE only <code>phase9ProductionPath.js:155</code>
<code>BatteryState</code>	<code>soh</code> , <code>internalResistanceMilliOhm</code> , <code>socThroughput</code> , <code>cycleCount</code> , <code>kappa</code> <code>@default(1)</code> , <code>kappaSampleCount</code> , <code>lastObservedSoc</code>	NONE ON THE LIVE PATH <code>energyCalibration.worker.js:118</code> is DEFERRED
<code>Robot.massKg</code>	vehicle mass, kg	ROBOTSPECIFICATION corrects the boundary docs
ambient / pack temperature	no column anywhere	NONE

The consequences, stated exactly

No candidate can be feasible today

`agentSnapshotLoaderFor` 's own comment: *"an agent with no `BatteryState` row carries `soc: null`, and F7/F34 refuse it."* No production path creates the row. Every agent, physical or simulated, is in that state.

No plan can have an energy

`consumption.legEnergyWh` → `{ ok: false, missing: ["energyModelParams"] }`.
Commissioning does not create one.

6000 mAh is not an energy

`packNominalWh` is watt-hours; `BOUNDS.batteryCapacityWh` is 1-200 000 Wh.
Converting needs a nominal pack voltage **nobody has declared**. At 12 V → 72 Wh; 24 V → 144 Wh; 48 V → 288 Wh. It sets charge power, reserve arithmetic, and every energy feasibility verdict.

The β coefficients are a *fitted physical model*

Supplying them asserts a measured fit. A simulation has a legitimate route — declare the β set as the simulated vehicle's *definition* and share the module — but then prediction and consumption agree **by construction, not by evidence**.

`kappa Float @default(1)` is the trap Step 5 documented

`coordinatorSolvePath.js:420-424` : *"a κ silently read as 1 is an uncalibrated fleet reported as a calibrated one."* A producer must write `kappa: 1` with `kappaSampleCount: 0` — the schema's own distinguishing mechanism — and never from telemetry.

`residualCv` is `null` by declaration

The schema says why: *"a fabricated dispersion produces three fabricated tier probabilities and F34 admits or rejects on them."* Its register sibling is Safety-class. **No value is proposed here.**

07 Charging audit

Everything the engine needs to consume charging is built and validated at the boundary. What is missing is the publisher.

COMPONENT	STATE
Charger	Full schema incl. <code>cellId</code> (the reachability-cache key), <code>chargerClass</code> , <code>ratedPowerW</code> , <code>lat/lon</code> , <code>isDepot</code> . Zero rows; no seed; no create path
ChargerReservation	Full schema incl. <code>targetSoc</code> , <code>projectionVersion</code> . Owned by the Scheduler (§14.7); the engine never writes one. No writer anywhere
ChargerAvailabilityProjection	<code>version @unique</code> , immutable by design — <i>"a projection that could be edited after a round consumed it would make that round unreplayable."</i> No publisher
<code>energy/chargeCurve.js</code>	§14.6's nonlinear CC/CV integrator. Already shared with VirtualRobot — the one place this programme got model-sharing right
<code>chargingSchedulerClient.js</code>	Complete engine half of §14.7: <code>resolveTargetSoc</code> , <code>assertNotEngineComputed</code> , <code>consumeProjection</code> , <code>unavailabilityEnvelope</code>
<code>chargerCandidatesFor</code> / <code>pinnedChargerProjection</code>	Built (E-11 / R-6). Nearest-first, truncated at <code>route.charger_reachability_k</code> , omits-and-names a charger with no <code>cellId</code>
<code>chargingStatusFor</code>	DOES NOT EXIST the classifier <code>indexMaintainer</code> needs
Simulator charging	Adopts the offer's <code>targetSoc</code> , integrates the shared curve, floors at present SoC. Charges in place — no station, no location, no slots, no queue. Pack temperature fixed at 20 °C

The owner's specification — one charger near MBA Block, three plugs, FIFO waiting, a realistic curve, and *"the robot does not have to reach 100 %; the Assignment Engine's energy validation decides"* — is a **conformant Charging Scheduler specification, near verbatim**. The last clause is especially good: it is precisely §14.6/§14.7's division of authority.

7.1 The 5-minute charge, and how to honour it without breaking §24.4

Bias beyond `sim.max_optimistic_bias` on any safety-relevant model — energy, charge, failure — fails the simulator's own release gate... "optimistic" means faster for travel time and less for energy.

`tools/simFidelity/validate.js` — §24.4's one-sided gate

A 5-minute full charge is optimistic on charge. Two resolutions; one is honest.

- Do not edit `CHARGE_POWER_CURVE` 's shape. That desynchronises the simulator from the module the server plans with — the exact coupling `constants.js` was written to preserve.
- Declare a high-power charger against a genuinely small pack. At 144 Wh, $0 \rightarrow 100\%$ in five minutes needs ≈ 1.7 kW average — high, but a real specification, and `Charger.ratedPowerW` is the column that states it. The five minutes then *follows from two declared physical facts* instead of overriding the model. If the declared voltage makes five minutes require an implausible power, that is the owner's number to revise, not the model's.

08 Telemetry / position audit

Complete from the wire to `Observation`. Broken from `Observation` to the engine.

```
VirtualRobot._emitTelemetry —emits {lat,lon,battery,speed,status
  → sockets/handlers/telemetry.handler.js
    → zod schema (passthrough), rate limit
    → §23.2/§23.5: a capability claim rejects the WHOLE frame
    → trustBoundaries.validatePosition / validateEnergy [R
    → persistentImplausibility → quarantine + security event
    → Redis live state · Robot row · Telemetry history
    → positionObservation.recordPositionObservation [S
      · requires the agent's own `timestamp`; NEVER substitut
      · sequence + observedAt high-water marks (replay refus
      · Observation{kind:"position", value:{lat,lon,provenan
```

★★ BREAK ★★ `indexMaintainer.worker` — DEFERRED, NOTHING START

```
→ AgentCellPosition (empty in every deployment)
→ Redis availability index
  ← candidates/expansion.js:416 candidatesInFineCel
  ← coordinatorSolvePath.js:313 agentSnapshotLoader
```

Verified by exhaustive search: `applyPosition` / `positionRecord` appear only in `indexMaintainer.worker.js`. There is no second writer and no

fallback.

The two blockers, from the worker's own header

THE ONE THAT MATTERS — A WIDENING DEFAULT

With no `chargingStatusFor`, `assembleRecord` supplies `charging: false`, and `availabilityIndex.classify` files an agent parked on a charger as `IDLE_READY` — the largest partition, searched first at §6.3 tier 1. The worker's own contract two paragraphs up claims defaults *"only narrow an agent's partition, never widen it into `IDLE_READY`."* The contract and the code disagree, and the code is what runs.

`capabilityAndContainerClassesFor` is also absent, but defaults to `[]`, which only empties the §6.2 secondary index — narrowing, and genuinely safe. So Finding 1 reduces to one classifier, and §07 shows exactly where it comes from.

`Observation` is correctly designed for this milestone and needs no change: §2.7's `OBSERVATION_SOURCE` is a closed set of five, both agent kinds are `AGENT_REPORT`, and `isPhysicalEvidence()` is **strictly positive** — an unlabelled legacy row is *not* physical evidence.

09 Assignment Engine path audit

Traced by call graph, not by file existence.

```
POST /api/tasks/assign (JWT + real User row)
→ task.service.assignTask
  · parsePayloadDeclaration → PayloadSpec upsert (idempotent)
  · requirementSetFor({chassisType, payloadMassKg}) → Task.requirementSet
  · task.robotId is NO LONGER HONOURED and says so in `ignoredFor`
  · cutoverEnabled.describe → live===false → 503 ENGINE_NOT_LIVE
  · prisma.task.create(PENDING)
→ admitToRound: $transaction{ leg.upsert · QUEUED timer · slaDeadline
  then intake.admit → WorkQueue
—— request path ends: durable, survivable, owned ——

shardSupervisor → election.assertConsensusStore UNDECLARED → no
→ leaderWorkers.apply({mayRunRound})
  | outbox / reconciler / timer → started
```

```

└─ COMPOSERS.coordinator(context) → coordinatorSolvePath.crea
    ★★ 26 of 34 unresolved → EXTERNAL_DEPENDENCY_UNAVAILABLE

— with all 34 supplied, the path exists and is complete —
round.execute
├─ expandCandidates → candidatesInFineCell ★★ EMPTY (see
├─ evaluateExact → feasibility 38 predicates → planBuilder
├─ → cellPairCache.hopsFor → deps.route(parts
├─ cost/phi · minCostFlow.solve · planState.reserve
└─ commit — Commitment + Outbox in ONE transaction (§10.3.2 sto
→ outbox.worker → OFFER → VirtualRobot._respondToOffer
→ drive → TASK_COMPLETE → lifecycle supervision → decisionRecord

```

9.1 There is no second assignment algorithm — confirmed

`task.service.assignTask` 's own header: *"Public API — §3.4's request path.*

One path, no branch." `dispatch/offers.js` : *"It does not decide **which** agent is offered a Leg (Phase 10's round)." DTARO's Mapbox-matrix nearest-robot selection is legacy comms/registry, not an intake path.*

`assignmentProjection.service.js` is explicitly *"a projection... it decides nothing, runs only after the decision exists, and writes no row the engine reasons from."*

THE REAL RISK

The architectural rule in the brief is **already satisfied by the tree**. The risk is not that a second path exists — it is that this milestone creates one. §26's ordering and §24's list are how that is prevented.

REAL IN PRODUCTION COMPOSITION TODAY

1 Request · 2 Durable queue · 3 The coordinator's *refusal* · 10 Dispatch · 11 Lifecycle supervision

IMPLEMENTED, NOT IN REAL COMPOSITION

4 Candidate generation · 5 Routing · 6 Feasibility · 7 Pricing · 8 Solve · 9 Commit · 12 Decision recording

10 UI audit

A rich existing surface. One missing dimension, and it must not be folded into an existing one.

SURFACE	STATE	VERDICT
<code>MapControl.jsx</code> 927 lines	Mapbox Standard, real DEM terrain at 1.0 exaggeration, 3D buildings, camera presets WORLD→CAMPUS, <code>useRobotStream</code> , <code>selectedRouteProgress</code>	A · REUSE
<code>mapboxMarkers.js</code> 520 lines	Per-robot marker; <code>batteryColor</code> ; status glyphs ⚡ ⚠ ○ ; zoom/pitch/heading transforms; selection + hover	D · ONE NEW STATE
<code>simulationIdentity.js</code> + <code>RobotIdentity.jsx</code>	3-valued identity; runtime RUNNING / PERSISTED_NOT_RUNNING / UNKNOWN; never reads <code>isOnline</code> , asserted structurally	A · REUSE the precedent to mirror
<code>AppProvider.jsx</code>	<code>ROBOT_UPDATE(D)</code> , <code>ROBOT_SPECIFICATION_UPDATED</code> , <code>TASK_CREATED/ACCEPTED/UPDATED/ASSIGNED</code> , <code>OFFER_RESPONSE</code> , <code>ALERT_CREATED</code> , <code>REROUTE_ALERT</code>	A · REUSE
Reconnecting state	DOES NOT EXIST no client staleness clock; ⚠ means <code>ISSUES</code> — a battery/fault status, not a link state	B · MISSING
Charging queue / slot occupancy	DOES NOT EXIST	B · MISSING
Simulator start/stop/reset controls	Deliberately absent — a per-operator button on a fleet-wide route would claim a scope the backend lacks	F · DO NOT ADD YET

Folding a link fault into `ISSUES` would make a one-second radio dropout indistinguishable from a low battery. §20 specifies the third dimension.

11 Existing tests audit

182 suites / 7 809 tests / 0 failures / 0 skips. Two gaps explain why this milestone's findings will be new.

Distribution: `engine/` 141 · `unit/` 25 · `gates/` 8 · `chaos/` 3 · `scale/` 3 ·
`integration/` 2. Live-DB harnesses run against disposable PG 18.3 on port
55432: `verify:step2/3/4/5`, `verify:v1CorePath`, `verify:t104`,
`verify:v10`, `phase9LiveDatabase`, `phase9ProductionPath`.

GAP 1 · FIXTURES CONCEAL MISSING ROWS

No test asserts that a commissioned robot ends up with a `BatteryState` or `EnergyModelParams` row. Every energy test injects them as fixtures. This is the defect class Step 3 already met once: *"a leak produced by nobody adding a field cannot be tested with a fixture."*

GAP 2 · THE WORKER IS NEVER SCHEDULED IN A TEST

No test drives `indexMaintainer` as a scheduled worker. `candidatesAvailabilityIndex` tests the *pure* classifier; the two harnesses that run the worker drive it **explicitly, one sweep at a time**. A wrong classification under a five-second loop is not reachable from either.

12 Gap matrix

Every row carries its evidence. Scroll horizontally for the full classification.

AREA	CURRENT STATE	REUSABLE?	MISSING?
VirtualRobot protocol	Complete; shares <code>fencing</code> , <code>chargeCurve</code>	YES, wholly	—
SimulationEngine	Complete; N robots	YES	—
<code>simulationPolicy</code>	Complete, 3 call sites	YES	—
Simulated creation + spec	SUPER_ADMIN, 8 required fields	YES	class presets
<code>Robot.massKg</code>	Column exists; operator-supplied	YES	the <code>vehicleMassKgFor</code> producer
Simulated speed	<code>5.56 m/s</code> fleet-wide; <code>MobilityModel</code> unread	partly	per-robot read
Simulated energy	flat <code>%/tick</code> ; 1000 Wh fleet-wide	no	consumption via \$14.2

AREA	CURRENT STATE	REUSABLE?	MISSING?
<code>consumption.legEnergyWh</code>	Written; fail-closed on 7 inputs	YES	its inputs
<code>EnergyModelParams</code>	Column  ; no producer	schema yes	producer + declared β
<code>BatteryState</code>	Column  ; no live producer	schema yes	producer at commissioning
Ambient / pack temperature	no column, no producer	no	<code>environmentFor</code> source
<code>chargeCurve</code>	Written; already shared	YES	—
Charger / Reservation / Projection	Full schema; 0 rows, no writer	schema yes	the Scheduler publisher
<code>chargingSchedulerClient</code>	Complete + validated	YES	its publisher
<code>chargingStatusFor</code>	absent; default widens into <code>IDLE_READY</code>	no	the classifier
<code>capabilityAndContainerClassesFor</code>	absent; <code>[]</code> narrows — safe	no	the reader
<code>indexMaintainer.worker</code>	DEFERRED; sole writer of the index	YES	two classifiers, then start it
<code>positionObservation</code>	Complete, live, provenance-labelled	YES	—
\$23.5 trust boundaries	Complete, live	YES	—
<code>route(parts)</code> — 6 fields	2 of 6 from Mapbox	2 of 6	dev adapter
<code>climbM</code> / <code>descentM</code>	Backend none; frontend has real DEM	YES, the data	server-side sampling
<code>stopStartCycles</code>	none from any product	no	declared assumption
<code>travelSdSeconds</code>	none from any engine (N29)	no	declared assumption

AREA	CURRENT STATE	REUSABLE?	MISSING?
<code>timeBucket</code>	nothing in <code>src/</code>	no	declared with the source
<code>executionGeometry</code>	Complete; correct seam; pre-transaction	YES	—
<code>coordinatorSolvePath</code>	Complete; refuses totally	YES	nothing
<code>p_fail</code> , <code>route_hazard_cost</code>	no producer; <code>reliability/</code> is a <code>.gitkeep</code>	no	dev seams
Battery wear (\$14.4)	<code>stressCurves</code>  ; mission quantities none	half	mission-quantities producer
15 register calibration rows	all <code>null</code> , <code>UNCALIBRATED</code> ; 2 Safety-class	no	owner / Safety publish
Serviceable region (F33)	Seam built (R-5); no declaration ; every non-empty derivation is <code>containmentOverlapping</code> — REFUSED	seam yes	owner decision, no escalation target
S-5 cutover binding	Mechanism verified over HTTP	—	refused by V9 +
<code>shardModel.ensureShard</code>	Complete, tested, no caller	YES	a caller
Map markers + routes	Complete	YES	reconnect state, charging queue
<code>sim:fidelity</code> (\$24.4)	Gate exists; no study → <code>INVALID DATA FILE</code>	YES	nothing — leave

13 Exact files likely to change

Authoritative means the file is the single owner of a rule; changes there must preserve it exactly.

FILE	EXPECTED CHANGE	RISK	AUTHORITATIV
server.js :659-724	Add the 11 dev seams — each behind one explicit dev posture, each absent when it is off	HIGH	YES — P15-R2 E-8b access discipline
workers/registry.js	index_maintainer : DEFERRED → SCHEDULED ; rewrite blockedBy ; a real cadenceParameter or @structural with a reason	MEDIUM	YES — S-2 is a live stop condition
workers/indexMaintainer.worker.js	Accept the two classifiers; make the absent-chargingStatusFor default narrow , matching its own contract	HIGHEST	YES — §6.2's classification
NEW services/chargingScheduler/	Estate, 3 slots, FIFO, reservations, target SoC, immutable projection, chargingStatusFor	MEDIUM	new
NEW routing/development/	Mapbox + DEM + declared fields; refuses to load outside dev posture; full provenance	HIGH	new
NEW services/agentEnergyProvisioning	EnergyModelParams + BatteryState at commissioning; kappa: 1 with kappaSampleCount: 0 ; residualCv left null	MEDIUM	new
NEW simulation/simulatedEnvironment.js	Ambient 25–35 °C over simulated time; pack temp rising under load; pure, seeded	LOW	new
NEW simulation/faultInjection.js	Comms loss, stale telemetry, malformed telemetry — a closed set	MEDIUM	new
simulation/VirtualRobot.js	Read the commissioned spec; consume via §14.2; fix the 2-stop truncation	HIGH	YES — the protocol conformance fixture.

FILE	EXPECTED CHANGE	RISK	AUTHORITATIV
			Mutation-tes every edit
simulation/constants.js	Fleet-wide values become documented <i>fallbacks</i> , never the answer when a row states one	LOW	partly
simulation/simulationConfig.js	Add fault controls to <code>LIVE</code> ; keep <code>PROTECTED_FIELDS</code> 'posture; never name <code>simulationOwnerId</code>	MEDIUM	YES — the boundary tes enforces the name ban
services/robotSpecification.js	LIGHT/STANDARD/HEAVY presets as data ; the mapping table untouched	LOW	YES — the on mapping
workers/coordinatorPipeline.js	Text only. Correct the stale <code>masses.vehicleMassKg</code> <code>why</code> . No probe, class or count change	LOW	YES — the contract
NEW Frontend lib/linkHealth.js	Pure staleness rules; 3-valued, mirroring <code>simulationIdentity.js</code>	LOW	new
Frontend mapboxMarkers.js	A link-health visual, independent of <code>status</code>	LOW	partly
NEW tools/verify/devSimulationE2E.js	Disposable PG 18.3 on 55432, real <code>server.js</code> , <code>real HTTP</code> <code>real sockets</code>	MEDIUM	new

14 Proposed architecture

One assignment path. One robot model. One telemetry pipeline. One routing contract. Three new producers and one new external service, each at a seam the repository already declares.

OWNER, once, in a decision record
DEVELOPMENT declarations: 3 class specs · pack voltage · β set · charger + rated power · return-leg Wh/m · spread model · stop

```

    · timeBucket rule · region cells · 15 register rows (2 Safety)
      | published, pinned ConfigVersion
      ▼
OPERATOR – POST /api/simulator/robot xN — (SUPER_ADMIN, 1 per 1
  | preset LIGHT|STANDARD|HEAVY prefills the 8 existing fields
  ▼
robot.service.createRobotWithProjection [UNCHANGED]
  | Robot{simulated, massKg, batteryReservePct, battery=SOC}
  | Agent (1:1) · AgentClass · MobilityModel · EnergyModel
  | ContainerModel · CapabilityBundle
  | NEW agentEnergyProvisioning: EnergyModelParams(β..., residualC
    BatteryState(soc, kappa=1, sampl
    ▼
SimulationEngine.addRobot → simulationPolicy → VirtualRobot [UNC
  | ★ reads ITS OWN row: speed ← MobilityModel, pack Wh ← Energy
  | mass ← Robot.massKg, SOC ← Robot.battery
  | ★ consumes via engine/energy/consumption.legEnergyWh (SHAR
  | ★ charges via engine/energy/chargeCurve (ALREADY SH
  | ★ simulatedEnvironment → ambientC / packC
  | ★ faultInjection → drops, staleness, garbage
  ▼ TELEMETRY
telemetry.handler → $23.5 → positionObservation → Observation(pos:
  ▼
★ indexMaintainer.worker (DEFERRED → SCHEDULED, 5 s)
  | chargingStatusFor ← ★ NEW Simulated Charging Schedule
  | capabilityAndContainerClassesFor ← CapabilityBundle / Con
  ▼
AgentCellPosition + Redis availability index ← the break this mi
  ▲
  |
  | candidates/expansion.candidatesInFineC
  | coordinatorSolvePath.agentSnapshotLoaderFor

OPERATOR – POST /api/tasks/assign (payload, NORMAL|HIGH|URGENT, S
  ▼
task.service.assignTask [UNCHANGED] → cutover gate → Task → WorkQu
  ▼
COMPOSERS.coordinator → coordinatorSolvePath.create(context)
  | 34 declared inputs: 11 seams from server.js, 15 from the reg
  ▼
round.execute → expand → feasibility → dev route() → phi → solve
  | commit — Commitment + Outbox in ONE transaction
  | executionGeometry resolved BEFORE the transaction
  ▼
outbox.worker → OFFER → VirtualRobot._respondToOffer
  | adopts targetSoc + reserves → assesses energy → assessExecuta
  | ★ walks ALL stops, not stops[0..1]
  ▼ ACCEPT | REJECT(→ feasibility observation) | DEFER(charging)
drive → TASK_COMPLETE → lifecycle supervision → decisionRecord

```

Every arrow above is an existing call except the five marked ★. No new database, no second robot model, no second routing contract, no second

telemetry system, no second assignment algorithm.

14.1 The Simulated Charging Scheduler

It runs **outside** `src/engine/**` (§1.6) and the engine only ever reads its output. Why this is the correct home and not a shortcut:

- `chargingSchedulerClient.js` is already the engine's validated half of this exact contract; **nothing in the engine changes**.
- `targetSoc` is the Scheduler's field. `assertNotEngineComputed()` becomes reachable and meaningful.
- It supplies `chargingStatusFor` from the party that actually knows — closing §08's blocker in the **narrowing** direction, from real reservations rather than from a default.
- `chargerReachability.worker` 's input appears for the first time.
- `unavailabilityEnvelope()` 's three pre-declared degradations become testable by stopping the Scheduler — a real §5.2 dependency-degradation exercise.

15 Simulation-only data model

Every row is `source = SIMULATION`, `environment = DEVELOPMENT`, **not physical evidence**. Rows marked `UNDECLARED` are not supplied by the owner's decisions in this brief.

#	ASSUMPTION	VALUE	LANDS ON
1-3	Nominal speed, LIGHT / STANDARD / HEAVY	1.2 / 1.5 / 1.0 m/s	<code>speedModel.nc</code>
4	Max speed per class <code>UNDECLARED</code>	—	<code>kinematicLimit</code> required; refusal
5	Payload capacity	3 / 5 / 8 kg	<code>ContainerMode</code>
6	Vehicle mass	1 / 2 / 3 kg	<code>Robot.massKg</code>
7	Pack capacity	6000 mAh	—

#	ASSUMPTION	VALUE	LANDS ON
8	Nominal pack voltage UNDECLARED — required for mAh → Wh	—	EnergyModel.p
9	Initial SOC	uniform 30–100 %	Robot.battery
10	batteryReservePct UNDECLARED — <i>not</i> §14.5's stack	—	Robot.battery
11	β coefficient set UNDECLARED — §14.2's model, nine terms	—	EnergyModelPa
12	betaThermal curves UNDECLARED	—	EnergyModelPa
13	residualCv	left null	sibling of a Sat
14	kappa	1 with kappaSampleCount: 0	BatteryState
15	soh UNDECLARED — §14.1 makes it load-bearing	—	BatteryState.
16–17	Ambient 25–35 °C varying; pack = ambient + load rise	—	environmentFo
18–19	Charger near MBA Block; 3 plugs, FIFO	—	Charger + Sch
20	ratedPowerW UNDECLARED — set by rows 7+8 and the 5- min target	—	Charger.ratec
21	chargerClass	STANDARD	Charger.charg
22	returnLegEnergyWhPerMetre UNDECLARED — β_dist is not a substitute	—	chargerReacha
23	travelSdSeconds model UNDECLARED	—	cellPairCache

#	ASSUMPTION	VALUE	LANDS ON
24	stopStartCycles derivation UNDECLARED	—	plan/timeline
25	timeBucket rule UNDECLARED	—	cellPairCache
26	climbM / descentM	REAL MEASUREMENT MEASURED_THIRD_PARTY_DEM	route()
27	p_fail UNDECLARED — owner says no per-class difference, so one fleet value	—	cost/cRisk
28	route_hazard_cost UNDECLARED	—	pricing
29	Wear mission quantities UNDECLARED — socThroughput, dod, socMid, cRate	—	energy/wear
30	Fault rates UNDECLARED	—	faultInjectio
31	Fleet size	3 initial, no hard cap	—
32	Replication posture — <i>a statement of fact about the dev cluster</i>	SINGLE_PRIMARY_NO_AUTOMATIC_FAILOVER	election.asse

FOURTEEN ROWS ARE UNDECLARED

The owner's decisions in this brief supply roughly half of what the milestone needs. The rest must be declared, and **this report does not choose any of them.**

16 Routing strategy

Three tiers, kept visibly apart.

`distanceM` and `travelSeconds` from Mapbox Directions;
`climbM` / `descentM` from **Mapbox Terrain-DEM** sampled along the returned **polyline**, as $\sum \max(0, \Delta h)$ and $\sum \max(0, -\Delta h)$; drivable geometry from `executionGeometry` (unchanged, and staying off the decision path);
`speedMetresPerSecond` from `MobilityModel.speedModel.nominalSpeedMps` per unit.

B · SIMULATED OR DERIVED, DEVELOPMENT ONLY, EACH LABELLED

`travelSdSeconds`

A declared spread model with a named author. A declared `0` is accepted by `buildEntry` and silence is not — but a declared `0` asserts a certain ETA and prices `p_late` optimistically. Declare non-zero, or leave absent and let the entry be refused. Not this session's choice.

`stopStartCycles`

Derived from Mapbox `steps=true` maneuver counts (a real change to `mapbox.service.js` behaviour and cost) or a declared per-distance rate. Labelled `SIMULATION_DERIVED`.

`timeBucket`

A declared bucketing rule over the pinned decision time, recorded as the rule the source was queried under.

C · FUTURE PRODUCTION — EXPLICITLY NOT BUILT

Self-hosted engine selection and ADR (B1 Steps 1/3/4/5); R1 self-hosting evidence; D8's extract identity, vintage, refresh cadence and re-contraction budget; `src/engine/routing/client.js` and \$5.2's degradation ladder; real congestion modelling; spread calibration.

THE FOUR RULES THE DEV ADAPTER MUST OBEY

1. One module, under an explicit dev posture, that **fails to load** when the posture is absent. `routing/client.js` is not created.
2. Its header states, in the `executionGeometry` register: *this is not a production routing source; it does not discharge B1; it does not close `engine_decision_path_wired`; it makes no release evidence admissible.*

3. Every field carries its provenance — `MEASURED_PROVIDER` , `MEASURED_DEM` , `SIMULATION_DERIVED` , `DECLARED_ASSUMPTION` — into the decision record, so a replay can tell which numbers were measured.
4. No network call inside the commitment transaction.

17 Battery / energy strategy

One model, shared — the way charging already is.

1. **EnergyModelParams** at commissioning. The declared β set per class, `residualCv` left `null` , `modelVersion: 1` , deterministic id per unit.
2. **BatteryState** at commissioning. `lastObservedSoc` from the declared initial SOC, `kappa: 1` with `kappaSampleCount: 0` , `soh` declared. Never written from telemetry.
3. The simulator consumes through `consumption.legEnergyWh` . Per step it forms the §14.2 profile — distance, climb/descent, stop-start, vehicle mass, payload, move time, dwell, ambient and pack temperature — and subtracts the returned Wh from a watt-hour pack state. A missing input surfaces as a refusal, not a plausible number.
4. SoC becomes a projection of energy, not the state itself. `soc = usableWh / packNominalWh` , `soh` -aware via `energy/usable.js` . The percentage thresholds become *derived* display values.
5. **Thermal**. `simulatedEnvironment` produces ambient and pack temperature; `betaThermal` evaluates the declared curves at both. `CHARGE_PACK_TEMPERATURE_C = 20` becomes the live pack temperature, which makes `temperatureDerating` reachable for the first time.
6. The `environmentFor` seam reads the same module, so engine prediction and simulator consumption see the *same* temperature.
7. Reserves stay the engine's. §14.5's four watt-hour layers. `Robot.batteryReservePct` is the unit's hardware floor and is kept apart.
8. **Battery wear**. `stressCurves` is a declared column value already read (W-A5). The mission quantities can be computed *from the simulated mission the simulator is about to drive* — the first legitimate producer that has existed.

Once the simulator consumes through the same module the engine predicts with, **agreement is by construction**. This milestone proves the pipeline *executes and is internally consistent*. It proves **nothing** about whether the β set describes a real robot. `sim:fidelity` stays `UNVALIDATABLE`, and no energy claim from this milestone is physical evidence.

18 Charging strategy

Build the publisher §14.7 names. Change nothing in the engine.

1. **The estate.** One `Charger` row: `isDepot: true`, `chargerClass: "STANDARD"`, lat/lon near MBA Block, a `cellId` from `cells.cellForPoint(lat, lon, FINE)`, and a declared `ratedPowerW`. A charger with no `cellId` is *omitted and named* — never routed to at a guessed distance.
2. **Three plugs, FIFO.** Slot occupancy and an arrival-ordered queue live in the Scheduler. A fourth robot waits; the Scheduler publishes its `projectedFreeAtMs` from the **charge-curve integral** of the robots occupying slots, not from a fixed wait.
3. **Target SoC is the Scheduler's and only the Scheduler's.** The engine reads it or falls back to the *published* class default with a recorded degradation flag.
4. **The projection is immutable and versioned.** Published from the *previous completed round* — §14.5's construction that breaks the `E_return` circularity by design rather than by iteration, and what makes a round replayable.
5. **A realistic curve, honestly.** Keep `chargeCurve`'s shape and integrator untouched. **Do not reshape the curve and do not compress the clock.**
6. **Interruption is already correct on both sides.** The owner's "*may stop charging and accept if the engine says there is enough energy*" is §14.6's protocol, implemented, and needs no new mechanism.

7. `chargingStatusFor` comes from the Scheduler's own reservations — the only authority for it. This is the item that lets `indexMaintainer` start safely.
8. **Test the degradation.** Stop the Scheduler and assert the three pre-declared degradations: in-progress charges complete, no interruption permitted (T3), target SoC falls back with a recorded flag, `E_return` falls back to depot-only.

19 Fault injection strategy

Four controls added to the existing closed set. No more.

CONTROL	SEMANTICS	EXERCISES (ALREADY BUILT)
<code>commsLossProbability</code>	Per tick, per robot. On fire, close the transport (<code>serverSocket.conn</code>), not the namespace — <code>socket.io-client</code> deliberately does not retry from <code>io server disconnect</code> . Recovery 1–2 s	<code>socket.io</code> reconnection; <code>_handlerSocket</code> (Step 4 finding 1); <code>disconnect</code> → <code>isOnline</code> false; the staleness sweeper
<code>staleTelemetryProbability</code>	A frame whose <code>timestamp / sequence</code> do not advance	<code>STALE_SEQUENCE / STALE_OBSERVED_AT</code> ; §2.7 replay detection; §7.3 staleness
<code>malformedTelemetryProbability</code>	An implausible position or energy, or a capability claim	§23.5 validators; capability claim → whole frame rejected ; <code>persistentImplausibility</code> → quarantine + security event
<code>faultScope</code>	<code>ALL</code> or one <code>robotId</code>	—

1. **Per-robot, from the robot's own seeded sub-stream.** Step 4 finding 5: two robots were coupled through the global `Math.random()` — adding a third changed the second's speed trace. Faults need a **third** sub-stream.
2. **LIVE** parameters, not **INITIAL** — they describe what may happen next, not where a robot started.

3. **A fault must not be modelled by lying in telemetry about a healthy robot.** A comms loss is a closed socket. A malformed frame is a malformed frame. Anything else tests the handler against a fiction.
4. **No new control plane.** Whether fault controls should be per-owner is the *same* open decision the `start/stop/reset` routes already carry, and this milestone should not settle it silently.

20 UI behaviour

Link health is a third, independent dimension. It is not a `status`.

DIMENSION	SOURCE	RENDERING
Identity	<code>simulationIdentity.js</code> — simulated / physical / unknown	<code>RobotIdentity</code> — words , 4 surfaces (shipped)
Operational status	<code>Robot.status</code>	Existing marker glyphs    
Link health	NEW <code>lib/linkHealth.js</code> — pure, 3-valued from telemetry recency	NEW red warning badge above the marker

1. **Moving robot remains visible.** The marker is *never removed* on a link fault. Its last known position and heading persist; motion must **visibly stop** rather than dead-reckon indefinitely — an animated marker on dead telemetry is the same class of fabrication as `isOnline: true`, which this programme has removed twice.
2. **Red warning symbol above the robot.** A dedicated badge in the same DOM element `applyMarkerCamera` / `applyMarkerHeading` already transform, so it inherits zoom, pitch and heading without a second transform path. **Distinct from**  **ISSUES**.
3. **Return to normal after reconnection.** The badge clears on the first *accepted* frame — a frame refused by §23.5 is not a recovered link.
4. **UNKNOWN is not LIVE.** A robot with no telemetry yet shows the neutral third state, with a structural test that `linkHealth` never reads `isOnline`.
5. **Charging surface.** Slot occupancy, the FIFO queue with positions, per-robot target SoC and projected free time — read from the Scheduler's

published projection, never recomputed in the browser.

6. **Assignment visualisation.** Reuse `taskPathCacheRef` , `selectedRouteProgress` , `OFFER_RESPONSE` . Map Phase 3's finding stands as a guard: **routes must not be deleted on every telemetry tick**, and route progress must actually reach the browser.

7. **No simulator start/stop/reset buttons**, for Step 3's stated reason.

21 Test plan

Additive. No existing test is modified to accommodate new behaviour.

#	SUITE	ASSERTS
T1	<code>robotEnergyProvisioning</code>	Commissioning creates <code>EnergyModelParams</code> and <code>BatteryState</code> ; <code>kappa == 1</code> with <code>kappaSampleCount == 0</code> ; <code>residualCv == null</code> ; idempotent
T2	<code>virtualRobotSpecificationFidelity</code>	A robot commissioned at 1.2 m/s moves at 1.2 , not 5.56; three classes, three distinct traces; pack Wh and mass from their rows
T3	<code>virtualRobotEnergyModel</code>	Consumption via <code>legEnergyWh</code> ; heavier payload ⇒ more Wh; positive climb ⇒ more; higher ambient ⇒ the declared effect; a missing profile field refuses
T4	<code>chargingSchedulerPublisher</code>	3 slots; the 4th queues; FIFO order; reservations carry <code>targetSoc</code> ; the projection is insert-only and monotone; the engine never writes either
T5	<code>chargingStatusClassifier</code>	A charging agent classifies out of <code>IDLE_READY</code> ; absence narrows, never widens ; <code>interruptible</code> from the reservation
T6	<code>indexMaintainerScheduled</code>	An Observation reaches <code>AgentCellPosition</code> within one sweep; <code>observedAtMs</code> is the agent's clock and a second sweep does not refresh it; a charging agent is not offered work

#	SUITE	ASSERTS
T7	developmentRoutingAdapter	Six fields finite and ≥ 0 ; a provider failure yields no entry ; DEM-sourced climb/descent; every field carries provenance ; refuses to load outside dev posture
T8	simulatedEnvironment	Ambient within 25–35 °C; varies with simulated time; <code>packC $\neq$ ambientC</code> and rises under load; deterministic under a seed
T9	faultInjection	Comms loss closes the transport ; recovery in 1–2 s; no duplicate handler registration after reconnect (needs a <i>dispatching</i> socket double, not a recording one); stale frames refused; a capability claim rejects the whole frame
T10	multiStopOfferExecution	A 3-stop plan is driven in full ; the current <code>stops[0..1]</code> truncation fails this test
T11	linkHealth (frontend arch)	3-valued; never reads <code>isOnline</code> ; the badge is distinct from <code>ISSUES</code> ; the marker is not removed on a fault
T12	simulationBoundary	Re-run unchanged. The engine still cannot see the discriminator; <code>simulationOwnerId</code> named by <i>no</i> file under <code>src/simulation/</code>
T13	coordinatorPipelineRequirements	Extended, not re-pinned. With the 11 dev seams: exactly 15 missing, all <code>REGISTER_UNRESOLVED</code> — this audit's measurement, pinned

Mutation testing is mandatory, on the Step 4/5 precedent: build a mutant per fix, run it, and record a **survivor as a survivor** — a mutant reported killed when it survived is worse evidence than no mutation testing. Priority targets: T5's narrowing default, T6's `observedAtMs` provenance, T7's refusal on provider failure, T10's stop walk.

22 Verification plan

The negative results matter as much as the positive ones.

LEVEL	COMMAND	EXPECTED
Unit / integration	npm test	exit 0; 182 + N suites, no existing test lost
Build gates	npm run gates	7 PASS / 1 FAIL gate:composition , unchanged, not weakened
Register	npm run gate:params	PASS, both rules — every snapshot.resolve() name published
Calibration	npm run gate:calibration	Unchanged. No Safety-class parameter moved
Routing readiness	npm run routing:readiness	BLOCKED D1/D3/D8, exit 0. Unchanged
Simulator fidelity	npm run sim:fidelity	UNVALIDATABLE the correct, required result. Never PASS
Regression harnesses	verify:step2/3/4/5, t104	All green, unchanged
V1 S-6 harness	verify:v1CorePath	EXIT 1 unchanged. This file is not to be modified
Milestone harness	tools/verify/devSimulationE2E.js	EXIT 0 PG 18.3 on 55432 (never Neon/5432), real server.js , real JWT, real HTTP, real Socket.IO, REDIS_ENABLED=false forced
Formal	—	Nothing re-run, nothing touched. No .tla , no .cfg
Mutation	per §21	Every mutant built, run, recorded; every file restored and hash-verified

The fifteen live assertions of devSimulationE2E.js

- 3 simulated robots commissioned through the real endpoint, each with EnergyModelParams + BatteryState and distinct class specs.
- Each connects, AUTHs, and emits telemetry with timestamp + sequence .
- Observation rows with provenance: "SIMULATED" , and isPhysicalEvidence() returns false for every one.

4. `indexMaintainer` sweeps them into `AgentCellPosition` ;
`candidatesInFineCell` returns **all three**; `observedAtMs` is the *agent's* clock.
5. A published, pinned DEVELOPMENT `ConfigVersion` with the 15 register rows, the region cells, and `cutover.engine_enabled = true` at region scope — **labelled, with the V9 accommodation recorded as an accommodation.**
6. A leader elected against a *declared* `SINGLE_PRIMARY_NO_AUTOMATIC_FAILOVER` store.
7. `COMPOSERS.coordinator` returns a **started handle** — 0 of 34 unresolved.
8. `POST /api/tasks/assign` with a payload and an SLA → 2xx, `Task` PENDING, `WorkQueue` row.
9. A round runs: candidates > 0, ≥ 1 candidate passes all 38 \$7.5 predicates, $\emptyset$ computed, `minCostFlow` solves.
10. **Commitment** and **Outbox** written in the same transaction; a `Round` row; a per-Leg Tier-A record; `searchGapMilliCU` is **null** when unproven, **never "0"** (S-1).
11. `OFFER` delivered signed and fenced; a `VirtualRobot` accepts and drives; `TASK_COMPLETE` ; `Leg` terminal; `Task` closed.
12. A low-battery robot defers while charging, then accepts *after the engine's energy validation passes* — not after reaching 100 %.
13. A fourth robot queues FIFO behind three occupied slots.
14. A comms-loss fault: reconnection within 1–2 s, **no command applied twice** (`socketGeneration` unchanged proves socket.io reused the object).
15. A malformed frame is refused by \$23.5 and the position Observation is **not** written.

23 Risks / possible regressions

Sixteen, ordered by severity. Four are critical.

#	RISK	SEVERITY	MITIGATION
R1	<code>indexMaintainer</code> starting with a widening default. The naive fix	CRITICAL	Do not touch the default until the Scheduler supplies a real classifier.

#	RISK	SEVERITY	MITIGATION
	(<code>charging: true</code>) makes <code>classify</code> return <code>null</code> for <i>every</i> agent and falsifies the shipped P9-DB-9		Order §18 before §07 of the plan. Re-assert P9-DB-9 at 9/9
R2	The dev routing adapter leaks into production posture — §17 rule 3	CRITICAL	One module; refuses to load without the posture; a structural test that no <code>src/engine/**</code> file requires it; <code>routing:readiness</code> re-measured
R9	A network call inside the commitment transaction — a hanging provider becomes a store-wide stall on locked rows	CRITICAL	Resolve all routing before <code>\$transaction</code> opens
R6	S-6 appears met by dev evidence. The dev harness would literally satisfy S-6's text	CRITICAL	Do not modify <code>v1CorePath.js</code> . A separate harness with a separate name
R3	A declared <code>0</code> in <code>travelSdSeconds</code> or <code>stopStartCycles</code> passes <code>buildEntry</code> while asserting certainty and zero stop-start energy	HIGH	Declare non-zero, or leave absent and let the refusal stand. Never declare 0 to get past a guard
R4	VirtualRobot regression. 2 234 lines, the conformance fixture, 5 Step-4 fixes, 6 documented limitations	HIGH	Mutation-test every edit; re-run <code>verify:step4</code> at 41/41; never touch the socket-keyed handler record
R5	Shared-model self-validation read as fidelity	HIGH	§17's limit in every module header; <code>sim:fidelity</code> stays <code>UNVALIDATABLE</code> ; no B-P gate touched
R7	<code>gate:composition</code> turning green from a dev start. §17 rule 6 permits removal <i>when the worker starts</i> — but a dev start is not a production composability claim	HIGH	Leave the <code>UNCOMPOSABLE</code> row. Gate stays RED. Owner decision if otherwise
R12	<code>kappa: 1</code> written without its sample count = an uncalibrated fleet reported as calibrated	HIGH	<code>kappaSampleCount: 0</code> , asserted in T1
R14	Region over-assignment. The 3 fine cells are 22.8× the campus — 95.6 % non-campus ground would be serviceable	HIGH	Owner decision only. Mitigating fact: tasks are manually created, so stops are operator-controlled

#	RISK	SEVERITY	MITIGATION
R8	<code>simulationBoundary.test.js</code> tripping — Step 4's <code>simulationConfig.js</code> tripped it immediately, and the right fix was <i>not</i> widening the allow-list	MEDIUM	Run it first, not last. Never name <code>simulationOwnerId</code> under <code>src/simulation/</code>
R10	Two <code>SimulationEngine</code> s over one fleet break AUTH — <code>commission()</code> re-mints the session key	MEDIUM	One engine per process; the harness asserts it
R11	A test tick-driver with two clocks froze docking and §18.5 timers while position and battery advanced — it looked like a movement defect	MEDIUM	One clock per driver
R13	Mapbox cost and rate limits. <code>steps=true</code> plus DEM sampling per hop, on the decision path	MEDIUM	Memoise behind the existing <code>cellPairCache</code> key; sample DEM once per cell pair; measure the added call volume before enabling
R15	A fourth <code>docs/v1/</code> status document — §17 rule 13	MEDIUM	Update <code>V1_IMPLEMENTATION_CONTROL.md</code> ; create no competing status file
R16	Scope creep into Phase 16 — every §22.5 kill switch is thrown	MEDIUM	§24

24 Things that must not be changed

§17's fifteen permanent rules plus five specific to this milestone. None has an exception for "to get the build green."

1 `tools/verify/v1CorePath.js`

—
the
instrument
S-
6's
verdict
rests

on.
Not
edited,
not
extended,
not
given
dev
values.
The
milestone
gets
its
own
harness.

2 **UNCOMPOSABLE.coordinator**

and **checkCompositionRoot.js**

. *declaratively*

Verified

this

turn:

the

gate

reads

the

table

, so

starting

the

coordinator

does

not

move

it.

3 **src/engine/feasibility/****

— *UNKNOWN IS NOT PERMISSION*

all

38

predicates

, ADMIT_WITH_PENALTY

includ

at

and DENY_UNLESS_ENVELOPE

. **The F17 fixture must not be shortened.**

4 `src/engine/cost/**`, `src/engine/solve/**`

— round.js

in

partic

's phi.assertWearChargedOnce

search

gap

prove

and

. S-

1.

5 `energy/chargeCurve.js`

—

the

curve

shape

and

the

integrator.

6 `energy/reserves.js`

— Robot.batteryReservePct

\$14.5%

four

watt-

hour

layers

must
never
be
written
into
it.

7 Fail-closed direction, everywhere.

absent never becomes

false; an unassigned cell is not an out-of-area one; an unread exclusion set is not an empty one.

8 positionObservation's refusals.

No timestamp

⇒ isPhysicalEvidence()

no

Observation

never

one

stamp

with

receipt

time.

stays

strictly

positive.

9 Observation's vocabulary.

No kind

sixth

source

no

second

, no AgentCellPosition

discri

column

on

10 `simulationBoundary.test.js` 's allow-lists

—
tightened,
never
widened.

11 `simulationPolicy.js` as the sole decider

of
the
two
simulation
conditions.

12 `executionGeometry` 's absence rule

— `NO_EXECUTABLE_PATH`
no
straigh
line
substi
and

must
not
be
defeated.

13 **Safety-class parameters.**

`energy.model_residual_cv`
`energy.reserve_floor_wh`
`route.degraded_reserve_factor`
`energy.max_combined_conservatism`

No
value
is
proposed
in

this
report.

14 `formal/**` , `docs/phase15/**` , `docs/release-decisions/RD-*` ,
 `docs/history/` , the §24 gate table, the release evidence.

Phase
15
is
FROZEN.

15 `sim:fidelity` 's one-sided algebra.

A UNVALIDATABLE
pessir
bias
is
never
a
failure
an
unval
mode
is

 PASS
never

.

16 The §23.5 trust boundaries.

Faults
are
injected
as
real
faults,
never
as a
relaxed
validator.

17 `routing/client.js` , `stores/roles.js` , `deps/registry.js` ,
 `circuitBreaker.js`

—
do
not
create.
An
old
plan
naming
a
module
is
not
evidence
it
should
exist.

18 **Phase 16.**

Every
\$22.5
kill
switch
is
thrown.

19 **No ninth stop condition.**

A
newly
found
core-
path
defect
is
ordinary
outstanding
work.

20 **No unrelated refactoring**

,
including
"while
I
was
in
there"
cleanups
on
the
core
path.

25 Things that should remain blocked

Blocked is a stated fact here, not a failure.

ITEM	STATUS AFTER THIS MILESTONE	WHY
B1 — routing selection, benchmark, ADR, R1	BLOCKED	A declared dev source is not a procurement decision
B8 — 39 Safety-class parameters	BLOCKED	§22.3. A labelled dev accommodation is not a Safety decision
S-5 production cutover	BLOCKED	V9 (2.0125 > 1.6 on the register's own defaults) + S2 (47 Safety changes, quorum 2). FD-3 = NO
D1 — boundary, CRS, cover, governance	BLOCKED	§7.1.2's containment refusal; escalation OPEN WITH NO TARGET
B-P — invariants, fidelity, soak, shadow	BLOCKED	Each needs an operating fleet and an observation window. Never simulate
B-O — safety case, rollback rehearsal	BLOCKED	Filed attestations by named humans
B-M — TLC capacities 2/3, boundedness	BLOCKED AT 2 OF 6	Compute and acceptance

ITEM	STATUS AFTER THIS MILESTONE	WHY
X3 — no <code>TASK</code> timer producer	OPEN, NON-BLOCKING	§4.2 has no transition table
Physical robot evidence	ABSENT	No hardware
Physical charger evidence	ABSENT	<code>Charger</code> will hold a simulated row
V1 stop condition	3 OF 8	S-3 needs all 28 including 2 Safety-class
Phase 15 §24 release verdict	BLOCKED	Frozen and untouched

26 Implementation order

Ordered by dependency, then by what unblocks the most. Steps 1–5 are pure repository work with no owner input. Step 6 is the first owner gate.

#	STEP	DEPENDS ON	CLOSES
0	Correct the two stale claims. <code>Robot.massKg</code> <i>exists</i> . Text only	—	A false claim published by code
1	LIGHT / STANDARD / HEAVY presets. Data in <code>robotSpecification</code> ; prefilled and editable in the UI. No schema, no new endpoint, no new class system	owner rows 1–10	Three distinct commissioned units
2	<code>agentEnergyProvisioning</code> — <code>EnergyModelParams</code> + <code>BatteryState</code> at commissioning	rows 8, 11, 12, 15	F7/F34's null-SoC refusal; <code>missing:</code> <code>["energyModelParams"]</code>
3	Simulator reads its own specification. Per-robot speed, pack Wh, mass	1, 2	§3.1's silent 4.6× divergence
4	<code>simulatedEnvironment</code> + the <code>environmentFor</code> seam	—	S-3 row 7; <code>betaThermal</code> ; <code>temperatureDerating</code>

#	STEP	DEPENDS ON	CLOSES
5	Simulator consumes through <code>legEnergyWh</code> . Watt-hour pack state; SoC becomes a projection	2, 3, 4	The dynamic-energy requirement
6	The Simulated Charging Scheduler. Estate, 3 slots, FIFO, reservations, target SoC, immutable projection, <code>chargingStatusFor</code>	rows 18–21 OWNER	F34/F35; §08's blocker; reachability input
7	Start <code>indexMaintainer</code> . Both classifiers; fix the contract/code disagreement; <code>DEFERRED</code> → <code>SCHEDULED</code> + an honest cadence (S-2)	6	The second structural break. Candidates exist
8	Fix the multi-stop offer truncation	—	§3.2 — required before charging insertion is live
9	Fault injection — 4 controls, a third seeded sub-stream	3	The fault requirements
10	UI: <code>linkHealth</code> + the red badge; charging queue; assignment visualisation	6, 9	The UI requirements
11	The DEVELOPMENT routing adapter. One module, one posture flag, full provenance	rows 22–26 OWNER	S-3 rows 1–5
12	The five remaining <code>NO_PRODUCER</code> seams at <code>server.js</code>	2, 5, 6 + owner rows	S-3 rows 8–11, 14 → 26 → 15
13	OWNER: publish the DEVELOPMENT <code>ConfigVersion</code> — the 15 register rows (2 Safety), region cells, the labelled V9 accommodation, the S2 approver record, the region binding, pinned. Plus the shard row <i>through</i> <code>shardModel.ensureShard</code>	OWNER + SAFETY ONLY	15 → 0. No repository act substitutes
14	<code>devSimulationE2E.js</code> — the 15 live assertions	1–13	The milestone's own evidence
15	Document. Update the control document and the contract; a note stating what the milestone does <i>not</i> establish	14	S-8's standard

Steps 1–5, 7–10 and 12's code halves can proceed today. Steps 6, 11 and 13 each need an owner declaration first, and step 13 additionally needs Safety.

27 Definition of done

Objective and command-checkable. All of the following, or the milestone is not done.

POSITIVE — MUST HOLD

1. `devSimulationE2E.js` → **exit 0**, all 15 assertions passing on disposable PG 18.3 (port 55432) with a real `server.js`, real HTTP and real Socket.IO.
2. `npm test` → **exit 0**, no existing test lost, the added count exactly accounted for.
3. `verify:step2/3/4/5`, `verify:t104` → all green, unchanged.
4. `phase9LiveDatabase.js` → **9/9**, including P9-DB-9.
5. `requirements()` against the real dev context → **0 of 34 unresolved**, and the composer returns a **started handle**.
6. A durable `Commitment` *and* `Outbox` row in **one** transaction, with a `Round` row and a per-Leg Tier-A record; `gate:erasure` still reconstructs byte for byte.
7. A simulated robot accepts, drives the **full** stop sequence, and completes.
8. A low-battery robot defers, then accepts on the engine's validation below 100 %.
9. A fourth robot queues FIFO; occupancy never exceeds 3.
10. Three robots move at three **different, commissioned** speeds; consumption responds to payload, mass, climb and temperature.
11. A comms-loss fault recovers in 1–2 s with **no duplicated command application**; the marker stays visible with the badge and clears on the first accepted frame.
12. Every mutant run, every survivor recorded as a survivor; every file restored and hash-verified.
13. Source digest and file count recorded before and after.

NEGATIVE — MUST ALSO HOLD, AND THESE ARE NOT OPTIONAL

- 14. `npm run gates` → 7 PASS / 1 FAIL, unchanged and not weakened.
- 15. `routing:readiness` → BLOCKED, D1/D3/D8.
- 16. `sim:fidelity` → UNVALIDATABLE , never PASS .
- 17. `gate:calibration` → unchanged; no Safety-class parameter moved.
- 18. `v1CorePath.js` → exit 1, and the file is byte-identical.
- 19. `simulationBoundary.test.js` → green, allow-lists not widened.
- 20. V1 stop condition: still 3 of 8. No Observation with provenance: "PHYSICAL" exists. `formal/` , `docs/phase15/` , the §24 table and the release evidence untouched.

28 V1 stop-condition impact

Net: 3 of 8 before, 3 of 8 after. That is the correct outcome.

#	EFFECT	WHY	SHOULD THIS MILESTONE
S-1	POTENTIALLY AFFECTED — RE-VERIFY	The milestone injects 11 inputs at the seams E-8 made fail-closed. Every declared value must be a declaration, and every absence must stay a refusal	NO — must remain M 8 suites stay green
S-2	DIRECTLY AFFECTED	Step 7 changes <code>index_maintainer's</code> readiness, and its <code>cadenceNote</code> records that <code>index.sweep_interval</code> is <i>not</i> in the register	NO — must remain M the parameter or ma <code>@structural</code> with th Never publish a nam not resolve
S-3	UNCHANGED — NOT MET	A DEVELOPMENT declaration of some inputs is not S-3. Two register rows are Safety-class with no provisional path; the region declaration is refused	NO. A dev decision re <i>different</i> record and r presented as answer: 09-05-01
S-4	PARTIALLY REACHABLE — DO NOT CLAIM	The composer genuinely returns a started handle	NO. Leave the row, le RED. Owner decisior

#	EFFECT	WHY	SHOULD THIS MILESTONE
		— the first half. But the gate reads <code>UNCOMPOSABLE</code> declaratively and stays RED	and recorded as one
S-5	UNCHANGED — REFUSED	V9 and S2 are <i>refusals</i> , not untaken acts. A labelled verification publish is precededented, but binding <code>cutover.engine_enabled</code> is the owner deciding the engine is live for a region	NO. The dev binding DEVELOPMENT-scope not discharge produc
S-6	THE SHARPEST RISK IN THIS PLAN	The dev harness would satisfy S-6's <i>text</i> — but on a tree carrying DEVELOPMENT-declared inputs, two of whose register rows Safety has not decided	NO — a hard constraint <code>v1CorePath.js</code> is not the milestone ships in harness; S-6 stays NO
S-7	UNCHANGED — ONE HALF HOLDS	<code>gates</code> fails only at <code>gate:composition</code> , i.e. S-4 → S-3	NO. <code>npm test</code> stays 1
S-8	AFFECTED — THE HONEST ONE TO MOVE ON	A simulation that executes end to end is exactly the artefact a reader will over-read. S-8 requires the disclaimers to be current at the <i>released tree</i>	YES — extend it. Add evidence is not physical model cannot validate dev router does not c the charger is simulated <code>sim:fidelity</code> is UNV

The milestone's value is a working, honestly-labelled development pipeline and the closure of two structural breaks and three missing producers that no audit had found — **not a number on a scoreboard**.

29 Final recommendation

What to build, what not to, and what the next implementation prompt should carry.

Implement

Available today, needing no owner input: step 0 (the two stale claims), step 1 (class presets on the existing path), step 2 (the `EnergyModelParams` and `BatteryState` producers — the highest-value item that needs nothing from anyone), step 3 (the simulator reading its own specification), step 4 (simulated environment), step 5 (consumption through §14.2), step 7's `capabilityAndContainerClassesFor` half, step 8 (the multi-stop truncation — a real defect, fixable now), step 9 (fault injection), step 10 (the UI link-health dimension).

Needing one owner declaration first: step 6 (the Simulated Charging Scheduler), then step 7's remaining half (starting `indexMaintainer`), step 11 (the dev routing adapter), step 12 (the five remaining seams).

Do not implement

A terrain engine or elevation table. A second robot model, class system, or telemetry pipeline. A second assignment algorithm, or any bypass of `coordinatorSolvePath`, feasibility, or the candidate/offer/commit path. A straight-line or stub router. `routing/client.js`, `stores/roles.js`, `deps/registry.js`, `circuitBreaker.js`. A fleet-creation API. A reshaped `chargeCurve` or a compressed simulation clock. Any modification to `v1CorePath.js`, the gates, or `UNCOMPOSABLE`. Any fabricated Safety-class value. Any fourth `docs/v1/` status document. Phase 16.

Must wait

For a physical robot

`provenance: "PHYSICAL"` Observations. Real β coefficients and `residualCv`. Real `soh`, internal resistance and `kappa`. Real pack and ambient temperature from sensors and the BMS. Measured `p_fail`, stop-start cycles and travel-time spread. B-P entirely. **Never simulate any of these.**

For Safety

`energy.model_residual_cv` and `energy.reserve_floor_wh`. B8's decision on `route.degraded_reserve_factor` or `energy.max_combined_conservatism` so V9's 2.0125 falls under 1.6. §22.3's two-person rule. The safety case and rollback rehearsal attestations.

For real routing

B1 Steps 1/3/4/5 and the selection ADR. R1 self-hosting evidence. D8's extract identity, vintage, refresh cadence and re-contraction budget. A calibrated `travelSdSeconds` source. A real congestion model. §5.2's degradation ladder.

What the next implementation prompt should contain

It should implement **steps 0–5 and 8 only** — everything needing no owner declaration — and carry:

1. **This audit's measurement as its baseline:** 26 of 34 today, 15 of 34 with every code seam injected, all 15 `REGISTER_UNRESOLVED`. It should re-derive this, not trust it.
2. **The three structural findings** — `indexMaintainer` never started (and that the naive fix falsifies P9-DB-9); no producer for `BatteryState` or `EnergyModelParams`; the Charging Scheduler is the missing publisher.
3. **The declared scope boundary:** no `src/engine/**` file edited; `v1CorePath.js`, the gates, `UNCOMPOSABLE`, `formal/**` and `docs/phase15/**` untouched; `sim:fidelity` stays `UNVALIDATABLE` — verified by an `mtime` sweep, not a statement of intention.
4. **The seven undeclared §15 rows it needs** before it can finish step 2, and the instruction to **stop and ask** rather than choose any of them.
5. §24's twenty rules verbatim, and §17's fifteen.
6. **Mutation testing as a deliverable**, with survivors recorded as survivors.
7. **Live verification on disposable PG 18.3, port 55432**, `REDIS_ENABLED=false` forced — never Neon, never 5432, never the hosted Redis in `.env`.
8. **The five Step-4 traps:** one socket survives reconnection (a recording-only double cannot see duplicate handlers); one clock per tick-driver; one engine per fleet; close the transport, not the namespace; `simulationOwnerId` named by no file under `src/simulation/`.

A separate, later prompt should carry steps 6, 7 and 11–14, once the owner's declarations exist — because each injects a value into a fail-closed guard, and a guard fed a value nobody declared is worse than a guard that refuses.

One closing statement, because it is the thing most likely to be misread. This milestone makes the real Assignment Engine run end to end against simulated robots, and that is genuinely valuable: it will execute stages 4 through 9 and 12 against a live database for the first time in this programme's history, and every previous pass that did so found defects a green suite had missed.

What it will **not** do is make any claim about physical reality true. The simulator and the engine will share the consumption model, the routing source will be a labelled development adapter over a hosted third-party API, the charger will be a row nobody installed, and two of the fifteen calibration values it needs are Safety's to decide and nobody else's. **The pipeline will be proven. The physics will not be.** Both halves of that sentence belong in the release note.

TRUTH > GREEN